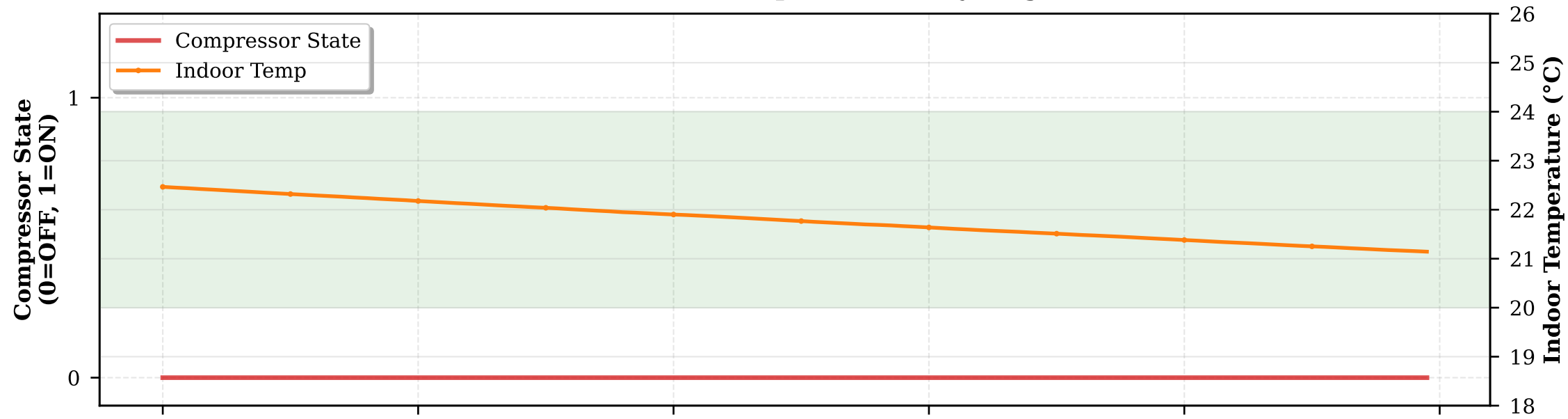


Figure 1: System Heartbeat - Prevention of Short-Cycling Damage

(a) Baseline Thermostat: Frequent Short-Cycling (0 switches)



(b) PI-DRL Agent: Stable Operation with Cycling Prevention (25 switches)

